CONSIDERATION OF OPTIMUM CEMENT CONTENT FOR MASS CONCRETE

A Memorandum by Roy W. Carlson, January 1981

The principal cause of cracking in massive concrete is the temperature change originating from the chemical action of the cement combining with water. If the concrete were free to expand or contract with temperature change, there would be no cracking. But there is nearly always some restraint either from rigid foundations or from one part of the structure changing in temperature more than another. It is almost axiomatic that the higher the temperature rise during hydration, the greater will be the temperature differences and the higher the temperature stresses.

It is only the temperature rise above a safe or tolerable value which causes serious cracking. This tolerable temperature is a few degrees above the "yearly average" or final temperature. A small increase in cement content above that which will cause the temperature to rise to the tolerable value can cause a large increase in the tendency to crack. In a typical example, an increase of 10 percent in cement content causes the amount of internal temperature rise above the safe value to increase by 20 to 30 percent. Conversely, A SMALL REDUCTION IN CEMENT CONTENT CAN CAUSE A LARGE REDUCTION IN THE TENDENCY TO CRACK.

It is obviously advantageous to precool the fresh concrete and to keep the cement content low enough to prevent cracking. This means using only enough cement to provide ample safety against crushing failure. For the mass concrete of Itaipu, there is no durability problem and no expectation of chemical attack; only enough compressive strength is required to support the expected stresses safely.

When design stresses are very high and dimensions large, cement content must be so high that cracking can not be prevented otherwise, joints should be used. If stress must be transferred across the joints, they should be gouted after cooling is complete. In the case of moderately thick walls, natural cooling may occur soon enough for the gouting. For more massive sections, embedded pipes should be installed for post cooling. However, the use of joints and pipe refrigeration requires much extra work, extra materials and extra cost. Therefore, every effort should be made to provide the necessary strength without the need for joints and without the danger of cracking.

It is customary to specify a strength of concrete which will provide an ample "factor of safety" against crushing. This factor is defined as the ratio of the compressive strength of concrete in the structure to the maximum design stress. An accepted procedure for determining the strength of concrete in the structure is to take the average strength of job cylinders and apply various corrections to compensate for the differences between the test cylinders and the mass concrete. The job cylinders are usually of 15 cm. diameter and 30 cm. length, wet-screened to remove all aggregate larger than 1.5 inch. The main correction is an empirical one called the "wet screening" correction. It is obtained by comparing the strength of the job cylinders with that of much larger test cylinders containing concrete exactly like that of the structure. Other corrections are made for "coefficient of variation", for "confidence factor", and for loss in strength due to long-continued loading. The sum of all these corrections is about 50 percent. This means that the job cylinders must withstand an ultimate stress about 50% higher than the product of safety factor and design stress. The need for modification of the correction will be discussed in what follows.

It may be noted that corrections to cylinder strengths are made for all unfavorable factors, but no corrections are made for factors which indicate the structure to be stronger than the job cylinders. A realistic approach whould be to consider both "minus" and "plus" correction factors. Let us consider how the current "minus" corrections might be modified and what "plus" corrections have been neglected.

1. The wet-screening correction (20 percent for Itaipu) is determined by test and most certainly should be applied without change.

2. The minus correction for variation in test cylinders has been taken as 15 percent, which is much too high. The strength of a massive dam is not determined by the weakest part, but more nearly by the average strength. This correction should be not over 5 percent.

3. The minus correction of 15 percent for "confidence factor" is not justified for Itaipu concrete because the inspection and testing are excellent. This should be zero.

4. The minus correction of 5 percent for loss in strength due to sustained loading is higher than found by long-time tests, but tolerable.

5. A small plus correction should be applied because of the shape of the structure being more favorable than that of the cylinders. If the test cylinders were to have a length-diameter ratio of 1.5 instead of 2.0, they would more nearly simulate the structure and they would be 3 percent stronger according to ASTM. This would be a plus correction of 3 percent.

6. A plus correction of at least 10 percent should be applied to allow for relaxation of overstressed regions. In the case of an arch dam, model tests show the strength of the arch to be double that indicated by comparing the strength of job cylinders with the maximum design stress. In the case of a buttress dam, a correction factor of 10 percent would be conservative.

7. A plus correction should be made because the compression tests do not reflect the true maximum stress in the test cylinder. When strains are measured on opposite sides of a cylinder being tested, there is always a difference, sometimes as much as 50 percent. Therefore, the ultimate stress in the cylinder is higher than the reported average. Since it is the maximum stress which is considered in the structure, it should be maximum stress in the test cylinders also. This would justify a plus correction of at least 5 percent.

The algebraic sum of all the corrections mentioned above is a minus 12 percent, or 8 percent less than the wet-screening correction alone. Therefore, if only the wet-screening correction were to be applied, this would appear to be conservative. And yet, it would allow a substantial reduction in strength requirement and consequent cement reduction. This is exactly what has been adopted for the Revelstoke Dam in Canada, only the wet-screening correction is being made.

At a meeting of Itaipu consultants in 1978, the late Prof. Nilo Amaral declared that the factor of safety against crushing of Itaipu concrete need not be more than 2.0. The writer is confident that if the factor of 2.0 were to be used, Itaipu Dam would be in no danger of failing by crushing and the dam would be free of temperature-caused cracks. However, there is no need to go so far.

The writer proposes that the factor of safety against crushing be kept at 3.0, which is what the U. S. Bureau of Reclamation uses. The only real change suggested by the writer is to reduce the correction factors to be applied to the strength of job cylinders to a single, conservative, and yet safe value much lower than the sum of factors now being applied.